



OPEN Functional characterization of dysregulated LncRNAs in BRCA and CESC highlights their role in cancer prognosis

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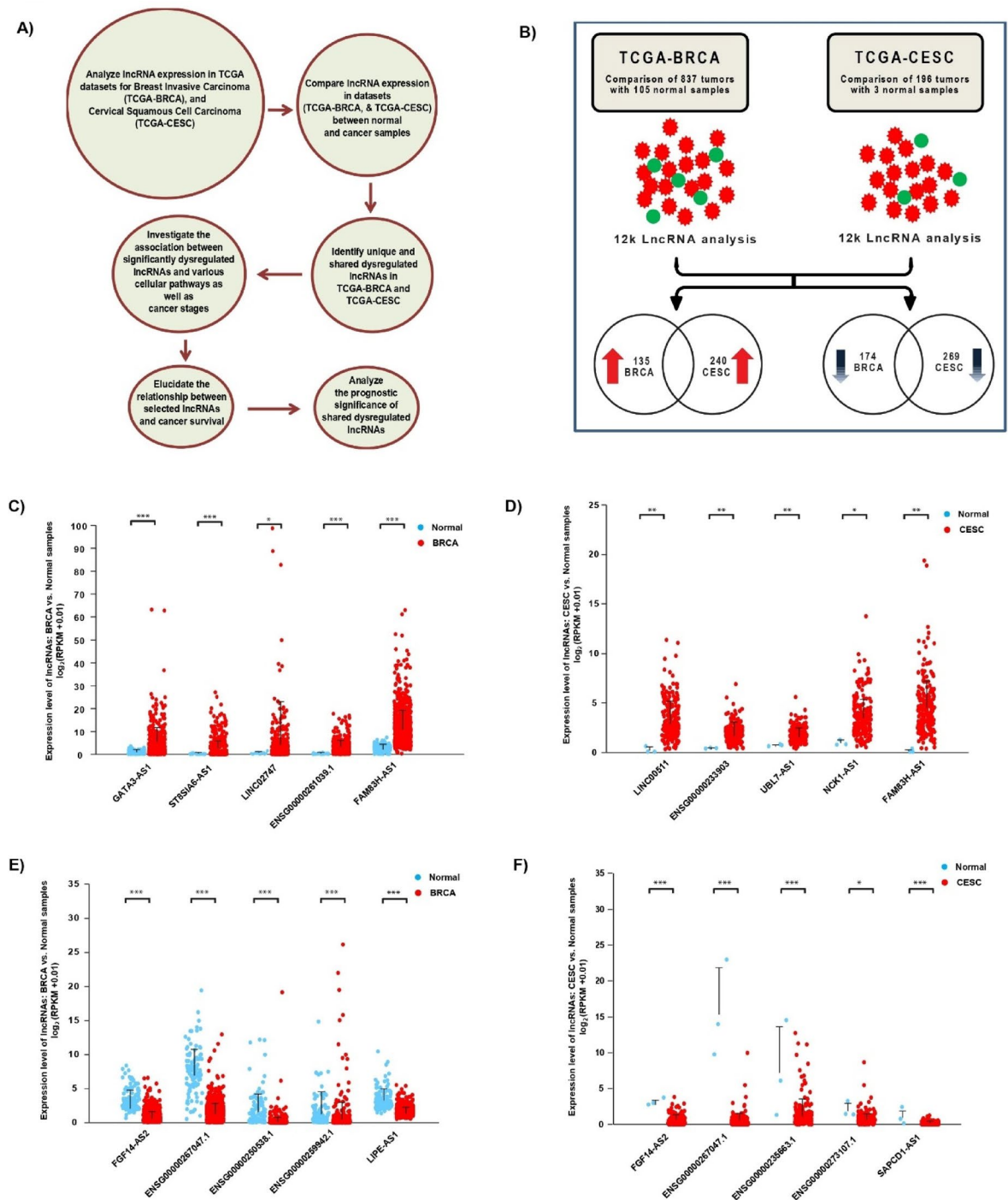
This study examines the contribution of dysregulated long non-coding RNAs (lncRNAs) to disease progression and prognosis in breast invasive carcinoma (BRCA) and cervical squamous cell carcinoma with endocervical adenocarcinoma (CESC) using data from The Cancer Genome Atlas (TCGA). Comparative analyses identified both shared and cancer-specific lncRNAs linked to tumorigenesis. Survival analysis using the Kaplan–Meier method highlighted several candidates with prognostic relevance. Notably, FAM83H-AS1 showed increased expression in advanced stages of BRCA and was linked with lymph node involvement and metastatic status. Its elevated expression was associated with poor survival outcomes in both BRCA and CESC cohorts. In contrast, lncRNAs such as FGF14-AS2, which were downregulated, demonstrated links to tumor-suppressive mechanisms and improved survival in BRCA patients. Subcellular localization analysis indicated predominant cytoplasmic presence of FAM83H-AS1, implying possible roles in post-transcriptional regulation. To support our in-silico observations, we carried out qRT-PCR experiments in breast cancer cell lines with varying degrees of aggressiveness, which confirmed the differential expression patterns of these lncRNAs. Overall, our findings highlight the clinical importance of lncRNA dysregulation in gynaecological cancers and point to their potential utility as biomarkers for prognosis and targeted therapy.

Keywords Long non-coding RNA (long cRNA, ncRNA), Prognostic biomarkers, BRCA and CESC, Cancer transcriptomics and functional enrichment analysis

The availability of large-scale transcriptomic databases such as The Cancer Genome Atlas (TCGA) and The Atlas of Noncoding RNAs in Cancer (TANRIC) has significantly improved our understanding of cancer biology^{1,2}. These platforms allow researchers to systematically analyse molecular alterations, including lncRNAs, which have been found to play important roles in cancer development and patient prognosis³. Despite these advancements, the exact functional roles of many lncRNAs in gynaecological cancers, particularly BRCA and CESC, are still not well understood.

Previous studies have applied bioinformatics-based approaches to explore the prognostic importance of lncRNAs. For example, HOTAIR was identified through TCGA and lncRNA-specific analysis platforms, showing its role in promoting metastasis and being linked to poor survival in breast cancer^{4,5}. Similarly, MALAT1, which was discovered through transcriptomic analyses, has been found to enhance migration and invasion in cervical cancer by regulating EMT and cell cycle-related genes^{6,7}. Another well-known lncRNA, MEG3, identified through differential expression studies, has been associated with p53 activation, apoptosis induction, and inhibition of cancer cell proliferation^{8,9}. Additionally, NEAT1, which was identified using expression datasets and clinical outcome correlations, has been implicated in tumor progression in various cancers, including gynaecological malignancies^{10,11}. These examples clearly highlight the importance of bioinformatics-driven studies in identifying functionally significant lncRNAs and their impact on cancer prognosis.

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Thus, lncRNAs are being explored for their potential clinical applications as biomarkers. For instance, HOTAIR and MALAT1 are being investigated as therapeutic targets and prognostic indicators in breast and cervical cancers^{12–14}. Dysregulation of these lncRNAs has been linked to chemotherapy resistance, making them critical factors in treatment planning^{15,16}. Similarly, MEG3 has been associated with better prognosis in various cancers, including gynaecological malignancies, due to its ability to activate p53-dependent apoptotic pathways and suppress tumor growth^{8,17}. On the other hand, NEAT1 has been reported to promote platinum-based chemotherapy resistance in ovarian cancer and is correlated with poor survival outcomes¹⁸. These studies emphasize how integrating bioinformatics tools like TANRIC with experimental validation can help discover clinically relevant lncRNAs, ultimately leading to better patient outcomes.

This study builds upon these findings by performing a systematic analysis of dysregulated lncRNAs in BRCA and CESC using TCGA and TANRIC datasets. We applied functional enrichment tools such as Enrichr and assessed subcellular localization using lncATLAS to identify key lncRNAs associated with patient prognosis^{19,20}. Among them, we found FAM83H-AS1, ENSG00000267047.1, and FGF14-AS2, which were linked to critical tumor-related pathways such as cell proliferation and apoptosis²¹. Kaplan-Meier survival analysis further confirmed their prognostic significance. Our findings not only provide functional insights into these lncRNAs

◀ **Fig. 1.** Transcriptome analysis identifies dysregulated lncRNAs in gynaecological cancers: TCGA-BRCA and TCGA-CESC datasets. **A**, The strategy for identifying dysregulated lncRNAs in gynaecological cancers, including BRCA and CESC involved analyzing their transcriptional profiles. **B**, Strategy for comparative analysis to reveal lncRNAs dysregulated in both BRCA and CESC datasets. Approximately 12,000 lncRNAs were analyzed, comparing expression between 837 tumor and 105 normal samples in the BRCA dataset, and 196 tumor and 3 normal samples in the CESC dataset. Upregulated lncRNAs were defined as having a fold change ≥ 2 (indicated by an upright arrow), while downregulated lncRNAs were defined as having a fold change ≤ 0.5 (indicated by an inverted arrow). Our analysis reveals that TCGA-BRCA and TCGA-CESC had 135 and 240 upregulated and 174 and 269 downregulated lncRNAs respectively. **C**, From the 135 upregulated lncRNAs in the BRCA dataset, 5 lncRNAs were prioritized based on statistical significance and prior associations with oncogenesis. The plot illustrates the relative expression levels of the selected lncRNAs: GATA3-AS1, STASIA6-AS1, LINC02747, ENSG00000261039.1, and FAM83H-AS1, comparing 837 tumor samples with 105 normal samples in the BRCA dataset. Each data point represents the expression level in a single sample, with the long and short horizontal bars indicating the mean and standard deviation (SD), respectively. Statistical significance determined using Student's t-test, demonstrates significantly altered expression in tumor samples (* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$). **D**, From the 240 upregulated lncRNAs identified in the CESC dataset, 5 lncRNAs were selected for further analysis. The plot depicts their relative expression levels, including LINC00511, ENSG00000233903, UBL7-AS1, NCK1-AS1, and FAM83H-AS1, comparing 196 tumor samples with 3 normal samples in the CESC dataset. Other details are the same as in part C. **E**, The graph represents the expression levels of the 5 selected lncRNAs from the BRCA dataset that were significantly downregulated in tumor samples. The analysis shows that the levels of FGF14-AS2, ENSG00000267047.1, ENSG00000250538.1, ENSG00000259942.1, and LIPE-AS1 are consistently lower in tumor samples compared to normal samples. Other details are the same as in part C. **F**, The graph highlights the expression of the 5 selected lncRNAs from the CESC dataset significantly downregulated in tumor samples. The levels FGF14-AS2, ENSG00000267047.1, ENSG00000235663.1, ENSG00000273107.1, and SAPCD1-AS1 were reduced in tumor samples compared to normal controls. Other details are the same as in part C.

but also emphasize their potential clinical utility as biomarkers and therapeutic targets, bridging the gap between molecular cancer research and clinical applications in gynaecological cancers.

Results

Dysregulated lncRNAs in TCGA-BRCA, TCGA-CESC

To explore lncRNAs with shared prognostic potential across female-specific cancers, we focused our primary analysis on BRCA and CESC. Although these cancers originate from different epithelial tissues, they exhibit several common biological and clinical features, including hormone responsiveness, derivation from reproductive organs, and high incidence among women worldwide. Previous studies have shown that lncRNAs such as HOTAIR, MALAT1, and MEG3 exert conserved regulatory roles in both breast and cervical cancers, affecting metastasis, proliferation, and resistance to chemotherapy via pathways like epithelial-mesenchymal transition and p53 signalling. The presence of shared molecular alterations in BRCA and CESC supports the idea that a combined transcriptomic analysis may help identify lncRNAs with broader significance in gynaecologic oncology. Furthermore, both TCGA datasets include adequate numbers of tumour and normal samples, allowing for meaningful statistical comparisons. This approach enabled us to identify lncRNAs that are commonly dysregulated in both cancers, some of which also showed prognostic relevance, thereby justifying our strategy of analysing BRCA and CESC together.

To elucidate the role of lncRNAs in gynaecological cancers, we carried out a comprehensive transcriptome analysis using TCGA datasets for TCGA-BRCA and TCGA-CESC. The workflow to identify dysregulated lncRNAs by comparing tumor and normal samples (Fig. 1A). In the TCGA-BRCA dataset, 837 tumor samples and 105 normal samples were analyzed, yielding 135 upregulated lncRNAs (fold change ≥ 2) and 174 downregulated lncRNAs (fold change ≤ 0.5) (Fig. 1B). Similarly, the TCGA-CESC dataset, consisting of 196 tumor samples and 3 normal samples, identified 240 upregulated and 269 downregulated lncRNAs. Comparative analysis of the two datasets highlighted lncRNAs shared among upregulated and downregulated genes across BRCA and CESC, signifying common oncogenic mechanisms. From the dysregulated lncRNAs, five upregulated candidates from each dataset were prioritized based on statistical significance, fold change, and known associations with oncogenesis. The expression levels of GATA3-AS1, STASIA6-AS1, LINC02747, ENSG00000261039.1, and FAM83H-AS1 were significantly elevated in TCGA-BRCA samples compared to normal tissues (Fig. 1C). Similarly, our analysis of CESC samples revealed five upregulated lncRNAs, LINC00511, ENSG00000233903, UBL7-AS1, NCK1-AS1, and FAM83H-AS1, which exhibited significant overexpression in tumor samples (Fig. 1D). Using the same approach, we identified downregulated lncRNAs in BRCA, including FGF14-AS2, ENSG00000267047.1, ENSG00000250538.1, ENSG00000259942.1, and LIPE-AS1 (Fig. 1E). Similarly, in CESC, downregulated lncRNAs such as FGF14-AS2, ENSG00000267047.1, ENSG00000235663.1, ENSG00000273107.1, and SAPCD1-AS1 exhibited significant reduction in expression (Fig. 1F), suggesting a potential role in maintaining normal cellular homeostasis, which becomes dysregulated during tumorigenesis. The identified lncRNAs potentially indicate their value as novel biomarkers with prognostic and therapeutic potential. Many of the above-identified dysregulated lncRNAs are known to be involved in pathways critical for cancer progression, including chromatin organization and resistance to apoptosis. Thus, we show that differential expression in tumor versus normal tissues offers an approach for biomarker identification.

Downregulated lncRNAs in BRCA and CESC links to tumor-suppressive pathways and poor survival outcomes

The Venn diagram illustrates the overlap between the BRCA and CESC datasets, identifying 68 shared downregulated lncRNAs (Fig. 2A). The observed downregulation suggests that these lncRNAs may play a vital role in preventing tumor growth, and their repression could contribute to the initiation and progression of cancer in both BRCA and CESC. Using RStudio 2023.12.0 Build 369, we drew a heatmap of the expression of these 68 lncRNAs downregulated in both BRCA and CESC tumor samples compared to normal controls (Fig. 2B). This shared repression pattern across two distinct gynaecological cancers suggests common regulatory mechanisms that, when disrupted, may drive tumorigenesis and affect patient outcomes. Further analysis revealed that only 49 of the 68 lncRNAs were significantly downregulated in both cancer types. All 49 lncRNAs, along with their fold-change magnitudes, are shown in Fig. 2C. Their average fold change indicates consistent and significant downregulation in both BRCA and CESC, supporting their potential role in cancer suppression.

High expression of downregulated lncRNAs in TCGA-BRCA tumors correlates with improved survival

We carried out the Kaplan-Meier survival analysis of 49 downregulated lncRNAs in the BRCA and CESC datasets and based on the data, we concluded that two ENSG00000267047.1 and FGF14-AS2 lncRNAs were significantly correlated to survival in BRCA cancer. lncRNA ENSG00000267047.1 expression had a significant

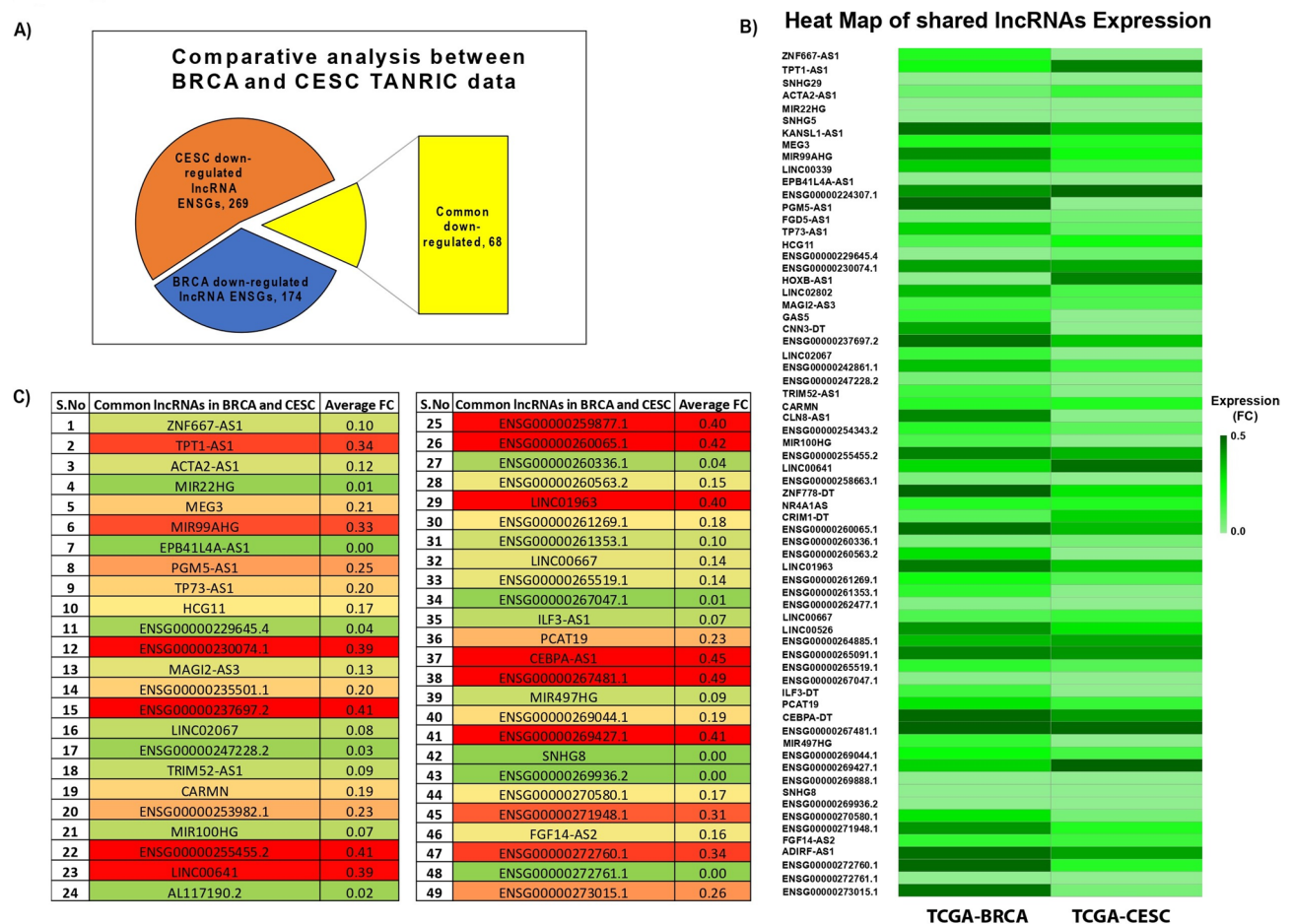


Fig. 2. Functional analysis of downregulated lncRNAs in BRCA and CESC tumors. **A**, Expression analysis identifies 68 lncRNAs downregulated in both the BRCA and CESC datasets. The analysis included approximately 12,000 lncRNAs, comparing expression between 837 tumor and 105 normal samples in the BRCA dataset, and 196 tumor and 3 normal samples in the CESC dataset. **B**, Heat map of lncRNA expression in tumors compared to normal controls illustrates the expression profiles of 68 lncRNAs that are downregulated in both the TCGA-BRCA and TCGA-CESC datasets. Generated using RStudio 2023.12.0 Build 369, each row corresponds to an individual lncRNA, while columns represent tumor samples from the BRCA and CESC datasets. This shared pattern of significant transcriptional repression suggests common regulatory mechanisms involving these lncRNAs in BRCA and CESC tumorigenesis. **C**, A total of 49 lncRNAs were found to be significantly downregulated in both BRCA and CESC datasets. The complete list of these lncRNAs is presented in the table, based on their consistent downregulation across both cancer types.

prognostic value in BRCA: Patients with high expression of ENSG00000267047.1 exhibit markedly better overall survival compared to those with low expression levels ($p = 0.0051$) (Fig. 3A). The survival curves clearly show a sustained survival advantage for the high-expression group throughout the observation period. This suggests that higher ENSG00000267047.1 expression may play a protective role in BRCA, possibly by modulating pathways critical for tumor suppression or cellular homeostasis. The consistent survival benefit across time emphasizes its potential as a robust prognostic biomarker in BRCA patients.

Patients with high expression of FGF14-AS2 in the BRCA dataset have significantly improved overall survival compared to the low-expression group ($p = 0.0004$) (Fig. 3B). The survival curves display the high-expression group maintaining a higher survival probability throughout the study. This early and sustained survival benefit indicates that FGF14-AS2 may be essential in regulating tumor progression or resistance to adverse clinical outcomes in BRCA. Its strong statistical association with better survival outcomes further supports its use as a potential prognostic marker, offering valuable insights into patient stratification and treatment planning in breast cancer. Taken together, our findings suggest that the combined evaluation of FGF14-AS2 and the previously uncharacterized ENSG00000267047.1 may serve as a novel and more effective prognostic indicator in breast cancer. Kaplan–Meier survival analysis in CESC patients showed no significant association for either ENSG00000267047.1 ($p = 0.9819$) or FGF14-AS2 ($p = 0.3431$), indicating that their prognostic relevance may be specific to BRCA (Supplementary Fig. 1). Neither lncRNA showed consistent or statistically significant differences with respect to pathological stage or nodal involvement. Although both were downregulated in tumor tissues compared to normal controls, their expression remained relatively stable across cancer stages, suggesting an early downregulation event rather than progressive suppression during disease advancement (Supplementary Fig. 2).

Functional analysis identifies the association of dysregulated lncRNAs with BRCA and CESC tumors

The overlap between upregulated lncRNAs of the BRCA and CESC datasets is illustrated by Venn diagram, which identifies 50 shared upregulated lncRNAs (Fig. 4A). We generated a heat map using RStudio 2023.12.0 Build 369 to display the expression of these 50 upregulated lncRNAs with TCGA-BRCA and TCGA-CESC tumor samples: a heatmap of 50 lncRNAs that are consistently upregulated in BRCA and CESC tumor samples compared to normal tissues (Fig. 4B). Out of 50 upregulated lncRNAs, only 10 were significantly upregulated in BRCA and CESC cancer types (Fig. 4C). The high expression levels of these lncRNAs in both datasets suggest their potential involvement in shared oncogenic pathways between BRCA and CESC. Next, we determined the cancer and tissue-specific expression of lncRNAs (Fig. 4D). FAM83H-AS1 is strongly associated with breast and colorectal cancers, while CRNDE is linked to malignant tumors of the cervix and metastatic neoplasms. Tissue-specific expression data show that these lncRNAs exhibit distinct patterns, with TMPO-AS1 being highly expressed in stromal cells, while CRNDE is prevalent in epithelial cells. Furthermore, protein interaction analysis for FAM83H-AS1 reveals connections with cancer-related proteins such as SPTBN1 and SPTBN2, which are involved in cell structure and signalling, indicating its potential role in tumorigenesis through these interactions. FAM83H-AS1 is associated with oncogenic signalling, making it a strong candidate for further investigation into its tumor-promoting functions. These shared lncRNAs underscore the presence of common molecular mechanisms driving tumorigenesis in these two gynaecological cancers. Localization analysis indicates that

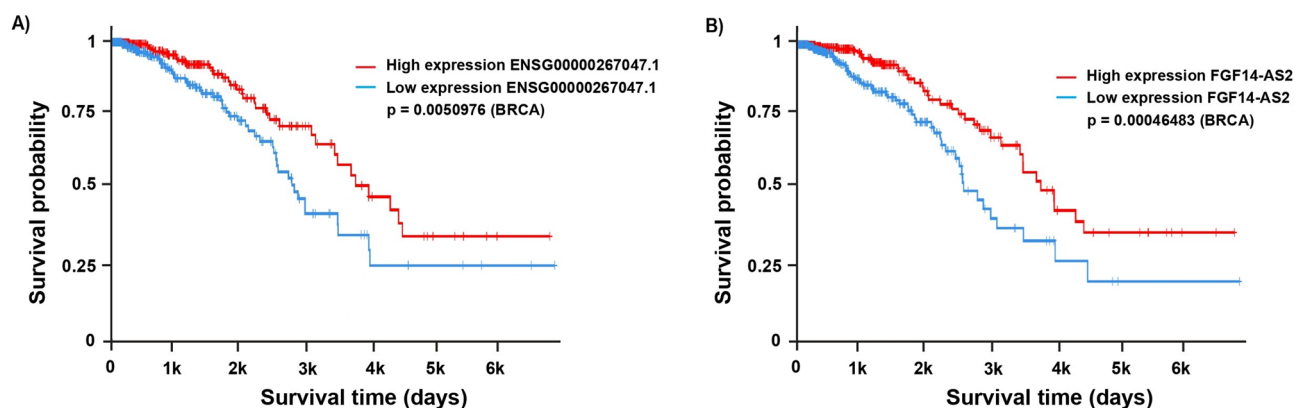
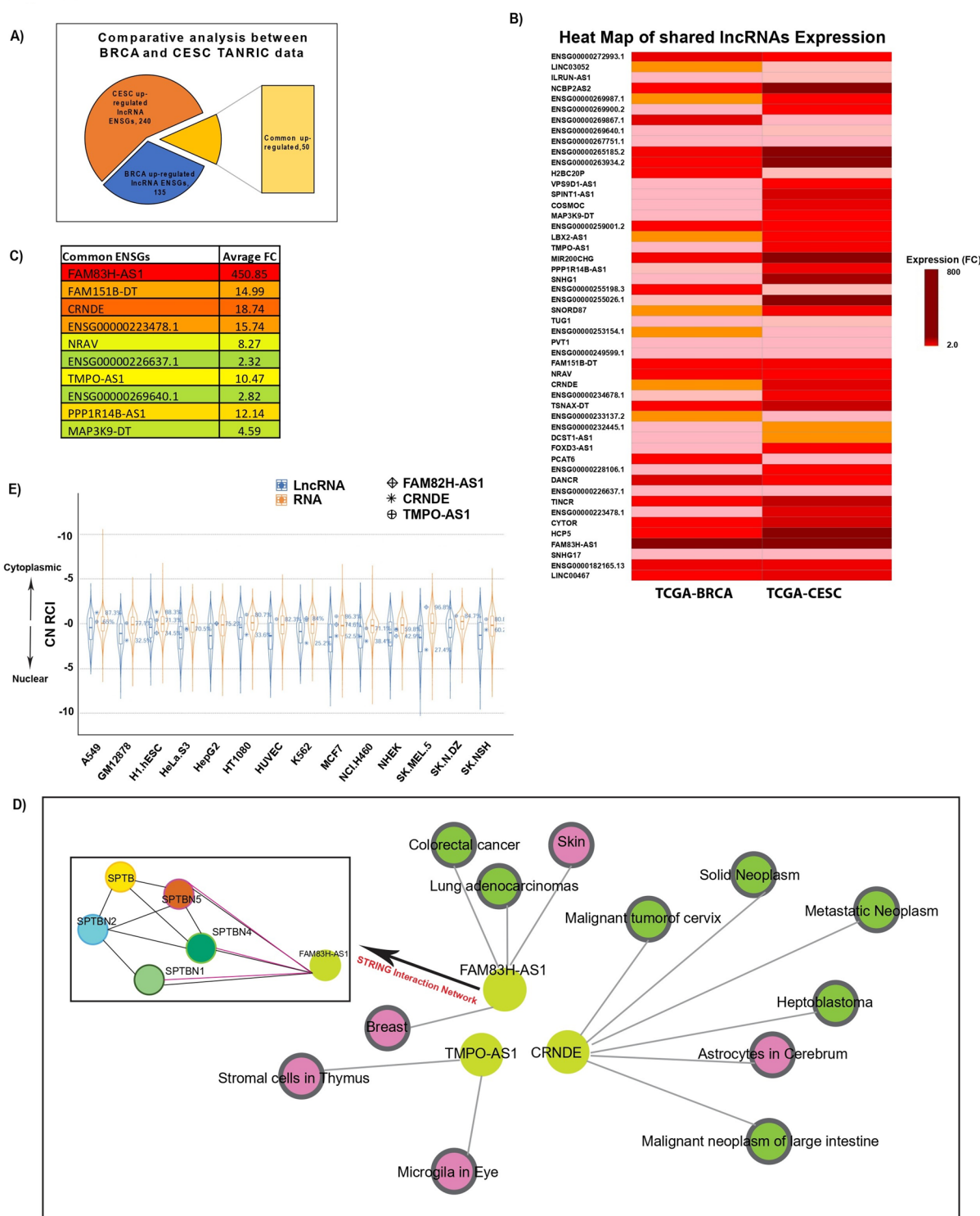


Fig. 3. Kaplan-Meier survival analysis displaying the estimated survival probability of BRCA patients. **A**, Kaplan-Meier survival curve comparing overall survival in BRCA patients with high versus low expression of ENSG00000267047.1. Patients with high expression of ENSG00000267047.1 had significantly better survival compared to those with low expression ($p = 0.0051$). Survival times are shown in days, emphasizing the prognostic relevance of ENSG00000267047.1 expression in BRCA. **B**, Kaplan-Meier survival curve comparing overall survival in BRCA patients with high versus low expression of FGF14-AS2. Patients exhibiting high expression levels demonstrate significantly better survival compared to those with low expression ($p = 0.0004$). Survival times are shown in days, emphasizing the prognostic relevance of FGF14-AS2 expression in BRCA.



FAM83H-AS1 and CRNDE are predominantly localized in the cytoplasm, suggesting their involvement in post-transcriptional regulation and signalling pathways (Fig. 4E).

FAM83H-AS1 expression correlates with breast cancer stages, nodal involvement, and metastasis

Clinical data used in this analysis were downloaded from the National Cancer Institute's Genomic Data Commons (GDC) data portal for TCGA BRCA, providing a robust and standardized dataset for evaluating the association between FAM83H-AS1 expression and breast cancer clinical parameters. The relative expression of FAM83H-AS1 was compared across pathological stages of breast cancer (Stage I, II, III, IV) (Fig. 5A). A significant increase in expression was observed between Stage I and Stage II ($p < 0.05$), as well as between Stage I and Stage III ($p < 0.01$), indicating a marked upregulation of FAM83H-AS1 expression in intermediate stages.

◀ **Fig. 4.** Functional analysis identifies the association of dysregulated lncRNAs with BRCA and CESC tumors. **A**, Expression analysis identifies 50 lncRNAs consistently upregulated in both the BRCA and CESC datasets. The analysis included approximately 12,000 lncRNAs, comparing expression between 837 tumor and 105 normal samples in the BRCA dataset, and 196 tumor and 3 normal samples in the CESC dataset. An intersecting Venn diagram represents 50 shared lncRNAs that are upregulated in both datasets compared to normal controls. **B**, Heatmap of differentially expressed lncRNAs in BRCA and CESC tumors relative to normal controls illustrates the expression profiles of 50 lncRNAs that are consistently upregulated in both TCGA-BRCA and TCGA-CESC datasets. The heat map was generated using RStudio 2023.12.0 Build 369, and it employs colour coding to represent normalized fold-change values. Each row corresponds to an individual lncRNA, while columns represent tumor samples from the BRCA and CESC datasets. These lncRNAs were selected based on upregulation compared to normal controls, with expression levels ranging from a 2-fold increase to 800-fold. **C**, A total of 10 lncRNAs were found to be significantly upregulated in both BRCA and CESC datasets. The complete list of these lncRNAs is presented in the table, based on their consistent upregulation across both cancer types. **D**, Functional enrichment analysis of upregulated lncRNAs identified in BRCA and CESC datasets using the Enrichr tool. The network highlights associations with key cancer-related pathways, gene ontology terms, disease networks, and cell-type-specific regulatory elements. Notably, FAM83H-AS1, CRNDE, and TMPO-AS1 demonstrate strong cancer associations. The identified network underscores the relevance of these lncRNAs in tumorigenic processes and highlights their potential as therapeutic targets or prognostic markers. **E**, This panel illustrates the cytoplasmic-nuclear localization patterns of the lncRNAs FAM83H-AS1, CRNDE, and TMPO-AS1 across various cell types. The Cytoplasmic-Nuclear Relative Concentration Index (CN RCI) is plotted on the y-axis, where positive values indicate cytoplasmic localization and negative values indicate nuclear localization. Each cell type is represented on the x-axis. Percentages represent the localization of the three lncRNAs. Violin plots also display the distribution of CN RCI values for lncRNAs (blue) and mRNAs (orange).

Although the expression appeared higher in Stage IV compared to Stage I, the difference was not statistically significant (ns), suggesting that while FAM83H-AS1 expression tends to increase in advanced stages, variability in expression prevents the change from being statistically validated. Similarly, when evaluating its expression across nodal involvement stages (N0, N1, N2, N3), no significant difference was observed between N0 and N1 (ns), indicating minimal change in the initial stage of nodal involvement (Fig. 5B). However, a statistically significant increase was observed between N0 and N2 ($p < 0.01$), highlighting a marked upregulation of FAM83H-AS1 in intermediate nodal metastasis. Interestingly, comparisons between N0 and N3 did not reach statistical significance (ns), despite an apparent increase in expression, suggesting variability in advanced nodal stages. FAM83H-AS1 expression did not show a statistically significant increase in N2/N3 compared to N1 in breast cancer (Supplementary Fig. 3).

Further, when comparing metastatic (M1) and non-metastatic (M0) cases across pathological stages, FAM83H-AS1 expression generally exhibited an upward trend with disease progression, particularly evident in Stage III. Although expression levels appeared lower in the metastatic Stage IV group compared to Stage III, this may be attributed to underlying biological variability. Notably, the mean expression values (depicted by dashed lines) closely reflected the pattern seen in Fig. 5A, reinforcing the association of FAM83H-AS1 with disease advancement (Fig. 5C). Analyzing FAM83H-AS1 expression across combinations of pathological stages and nodal involvement revealed that its expression was relatively low and consistent in early-stage cancers (Stage I and II) across all nodal involvement levels (Fig. 5D). However, in Stages III and IV, a trend of increased expression with higher nodal involvement (N2/N3) was observed, further supporting its association with cancer progression and nodal metastasis. Finally, when FAM83H-AS1 expression was analyzed across tumor size (T stages), a statistically significant increase was observed between T1 and T2 ($p < 0.05$) and between T1 and T4 ($p < 0.001$), suggesting a positive correlation between its expression and tumor size/invasiveness (Fig. 5E). Together, these findings highlight the potential role of FAM83H-AS1 in breast cancer progression, nodal metastasis, and tumor aggressiveness, particularly in intermediate and advanced stages.

FAM83H-AS1 is significantly upregulated in nodal metastasis but not in tumor progression in CESC

The expression analysis of FAM83H-AS1 in breast cancer revealed a significant correlation with tumor progression, nodal metastasis, and tumor size, with notable upregulation in intermediate pathological stages, early nodal involvement, and larger tumors. However, its expression appeared to plateau in advanced stages, suggesting a potential saturation effect or increased variability in highly aggressive cases. Building upon these findings, the analysis in CESC aimed to determine whether a similar pattern of FAM83H-AS1 regulation exists across different cancer types. We evaluated the relative expression of FAM83H-AS1 across different nodal stages in CESC samples. A significant upregulation was observed when comparing N0 to N1, suggesting that FAM83H-AS1 may be more closely linked to lymphatic spread in CESC rather than primary tumor growth (Fig. 6A). The box plot analysis further stratified FAM83H-AS1 expression across nodal involvement (N0, N1) and tumor stages, showing greater variability in expression, particularly in T1 cases but a distinct trend of increasing FAM83H-AS1 expression from T2 to T4 stages was observed across nodal stages (N0 and N1) (Fig. 6B). Together, these findings highlight a potential cancer association of FAM83H-AS1.

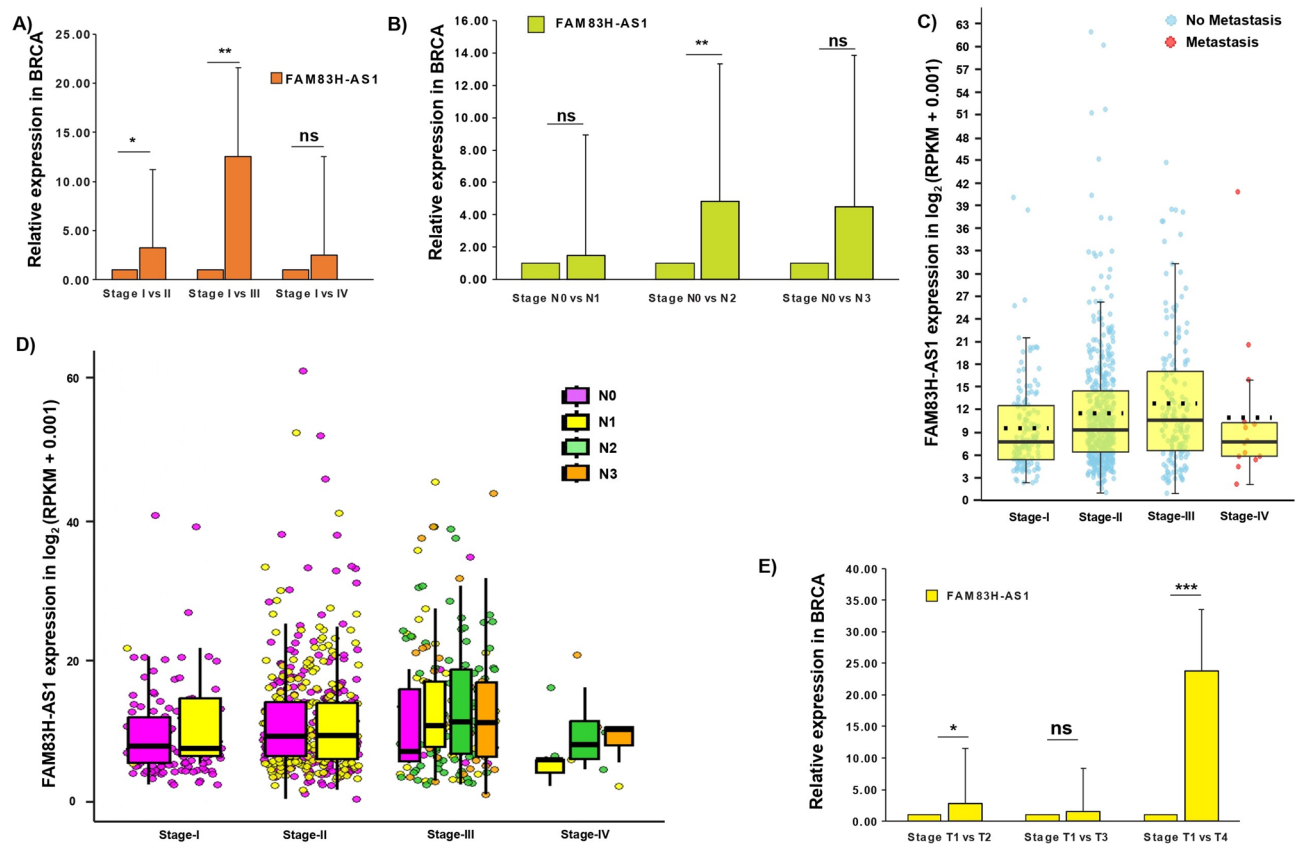


Fig. 5. Correlation of dysregulated lncRNA FAM83H-AS1 with disease progression, tumor burden, and metastatic potential in breast cancer. **A**, Relative expression of FAM83H-AS1 across overall pathological cancer stages. Expression was compared between different stages to evaluate its correlation with disease progression. The left vs. right bars represent the expression in the same sequence as the two stages labelled below. **B**, Relative expression of FAM83H-AS1 compared among different pathological nodal (N) stages. Expression was analyzed between Stage N0 (no lymph node involvement) and subsequent nodal stages (N1, N2, N3). **C**, Box plot showing log₂-transformed expression (RPKM + 0.001) of FAM83H-AS1 across pathological cancer stages (Stage I-IV) in breast cancer, stratified by metastatic status. Non-metastatic samples (M0) are shown in blue, and metastatic samples (M1) in red. Each box represents the interquartile range (IQR), with the solid line indicating the median and the dashed line indicating the mean expression. This plot includes 755 out of the 814 total breast cancer samples shown in part A while samples lacking definitive metastatic stage information were excluded. **D**, Box plot illustrating log₂-transformed expression (RPKM + 0.001) of FAM83H-AS1 across pathological cancer stages (I-IV) combinations and nodal involvement (N0-N3). Data points represent individual samples, while the boxes and whiskers represent group medians and variability. **E**, Relative expression of FAM83H-AS1 across pathological tumor phases (T1 vs. T2, T1 vs. T3, and T1 vs. T4). Tumor phase reflects the size and extent of the primary tumor. Data are presented as mean $\pm$ SD, and significant differences are indicated as $p < 0.001$ (***), $p < 0.01$ (**), $p < 0.05$ (*), and $p > 0.05$ (ns).

Prognostic relevance of FAM83H-AS1 in gynaecological cancer

The Kaplan-Meier survival analysis demonstrates that high expression levels of FAM83H-AS1 are associated with significantly reduced overall survival in BRCA patients (Fig. 7A). The BRCA cohort, consisting of tumor samples stratified by median expression levels of FAM83H-AS1, shows a clear distinction in survival outcomes. Patients in the high-expression group had a notably lower survival probability compared to those in the low-expression group ($p = 0.034149$). Survival times, measured in days, highlight a consistent trend where elevated FAM83H-AS1 expression correlates with poorer clinical outcomes, underscoring its potential as a negative prognostic marker in BRCA. This result indicates that FAM83H-AS1 could serve as a critical biomarker for predicting patient prognosis and guiding personalized treatment strategies in breast cancer.

Similarly, in CESC patients, Kaplan-Meier survival analysis reveals a significant association between FAM83H-AS1 expression levels and overall survival (Fig. 7B). The CESC cohort, divided into high and low expression groups using the median expression threshold, shows that higher expression of FAM83H-AS1 is linked to poorer survival outcomes ($p = 0.034426$). A striking feature of this analysis is the abrupt decline in survival probability for patients in the high-expression group within the first 1,000 days, whereas the low-

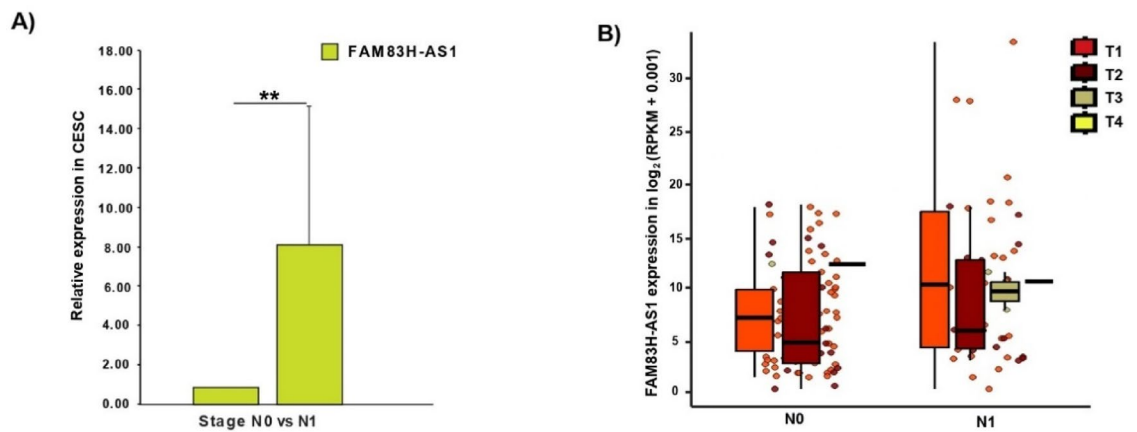


Fig. 6. Expression of FAM83H-AS1 in CESC across tumor stages and nodal involvement. **A**, Relative expression of FAM83H-AS1 in CESC between different nodal stages (N0 vs. N1). A significant upregulation of FAM83H-AS1 expression was observed in N1 compared to N0 ($p < 0.01$), suggesting a potential association between FAM83H-AS1 expression and lymph node metastasis in CESC. Data are presented as mean $\pm$ standard deviation (SD), with statistical significance indicated as $p < 0.01$ (**). **B**, Box plot showing $\log_2$ -transformed expression (RPKM + 0.001) of FAM83H-AS1 across nodal stages (N0 and N1) further categorized by tumor stages (T1, T2, T3, T4). While there is variability in expression, particularly in T1 cases, a distinct trend of increasing FAM83H-AS1 expression from T2 to T4 stages was observed across nodal stages (N0 and N1). All statistical comparisons were conducted to evaluate the potential role of FAM83H-AS1 as a biomarker in CESC.

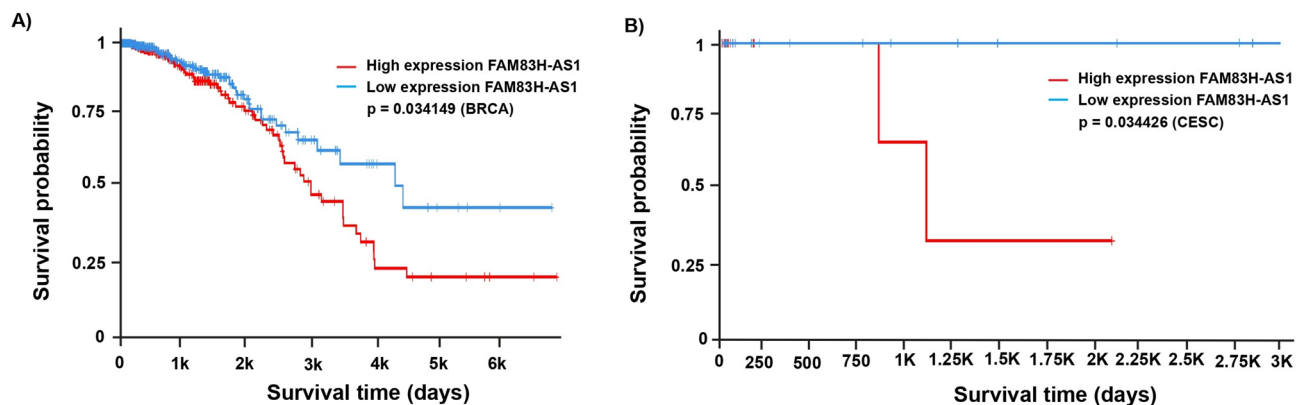


Fig. 7. Survival analysis of FAM83H-AS1 in BRCA and CESC patient groups. **A**, Kaplan-Meier survival analysis illustrating the correlation between FAM83H-AS1 expression levels and overall survival in BRCA patients. The BRCA cohort was stratified into high and low FAM83H-AS1 expression groups based on median expression values. Patients with higher FAM83H-AS1 expression exhibited significantly reduced survival probability compared to those with lower FAM83H-AS1 expression ($p = 0.034149$). Survival times are shown in days (k represents 1000), emphasizing the prognostic relevance of FAM83H-AS1 expression in BRCA. **B**, Kaplan-Meier survival analysis demonstrating the association between FAM83H-AS1 expression levels and overall survival in CESC patients. The CESC cohort was divided into high and low FAM83H-AS1 expression groups using median expression levels as the threshold. Higher FAM83H-AS1 expression was significantly associated with poorer survival outcomes ($p = 0.034426$), highlighting the potential prognostic value of FAM83H-AS1 expression in CESC.

expression group maintains a higher survival probability over the same period. This sharp separation between the two groups underscores the profound impact of FAM83H-AS1 expression on patient outcomes in CESC. This pattern is consistent with observations in BRCA, reinforcing the potential role of FAM83H-AS1 as a prognostic biomarker across multiple cancer types.

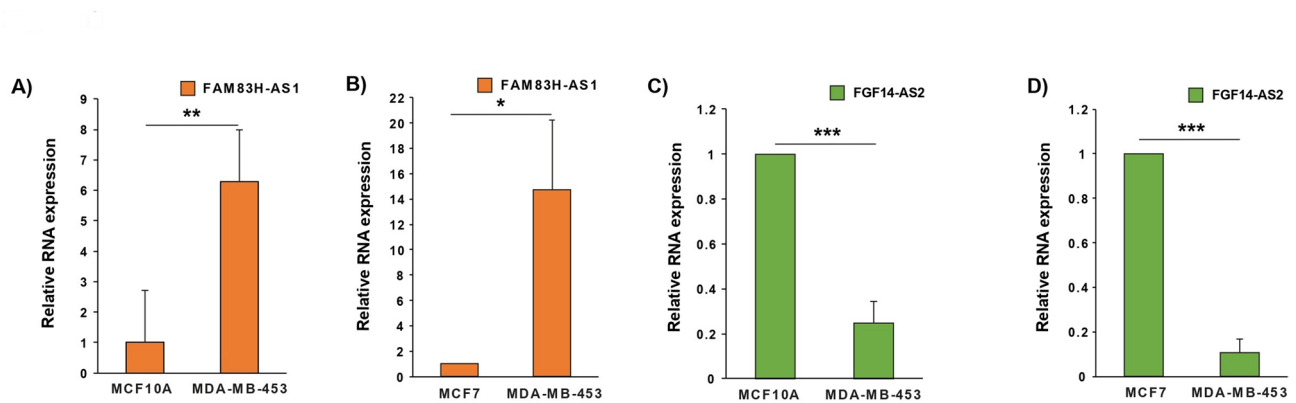


Fig. 8. Experimental validation of FAM83H-AS1 and FGF14-AS2 expression in breast cancer cell lines of differing aggressiveness. **A**, Expression of FAM83H-AS1 in the non-tumorigenic breast epithelial cell line MCF10A compared to the aggressive breast cancer cell line MDA-MB-453. **B**, Expression of FAM83H-AS1 in the less aggressive breast cancer cell line MCF7 versus MDA-MB-453. **C**, Expression of FGF14-AS2 in MCF10A and MDA-MB-453, representing non-tumorigenic and aggressive phenotypes, respectively. **D**, Expression of FGF14-AS2 in MCF7 and MDA-MB-453, evaluating expression changes associated with increasing tumor aggressiveness. All values represent mean $\pm$ standard deviation (SD) from three independent experiments. Statistical significance was assessed using Student's t-test: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Experimental validation of FAM83H-AS1 and FGF14-AS2 expression across breast cancer cell lines of varying aggressiveness

To experimentally validate the expression patterns of key prognostic lncRNAs identified in the TCGA-BRCA dataset, we assessed the expression of FAM83H-AS1 and FGF14-AS2 across a panel of breast cell lines representing varying levels of aggressiveness. MCF10A, MCF7, and MDA-MB-453 share an epithelial origin but differ in tumorigenic potential, with MCF10A being non-tumorigenic, MCF7 representing less aggressive luminal-type cancer, and MDA-MB-453 modelling a more aggressive HER2-positive phenotype. This panel offers a comparable cellular background while capturing a spectrum of breast cancer aggressiveness. FAM83H-AS1 expression was elevated in MDA-MB-453 cells relative to both MCF10A and MCF7 (Fig. 8A and B), whereas FGF14-AS2 expression was markedly reduced in MDA-MB-453 cells compared to the other two lines (Fig. 8C and D). These trends are consistent with our computational findings, which indicated upregulation of FAM83H-AS1 and downregulation of FGF14-AS2 in more aggressive breast tumors. By demonstrating differential expression of these lncRNAs across a phenotypic gradient of breast cancer cell lines, the qRT-PCR findings validate the computational data and suggest a functional association between lncRNA expression and breast cancer progression. Collectively, these results underscore the biological relevance of FAM83H-AS1 and FGF14-AS2 as potential prognostic biomarkers and candidate regulators of malignancy.

Comparative analysis of dysregulated lncRNAs across BRCA, CESC, and OV identifies a prognostic marker in ovarian cancer

To evaluate the relevance of lncRNAs identified in BRCA and CESC within ovarian cancer (OV), we extended our analysis to the TCGA-OV dataset. Due to the absence of normal controls and Stage I samples in this cohort, we compared Stage III/IV ($n = 388$) with Stage II ($n = 21$) tumors to identify lncRNAs associated with disease progression, resulting in the identification of 35 upregulated and 45 downregulated lncRNAs (Supplementary Fig. 4A). We then examined their overlap with the BRCA and CESC datasets. Although six lncRNAs were commonly upregulated, none achieved statistical significance across all three datasets based on t-test results, suggesting that their dysregulation may be cancer-type specific (Supplementary Fig. 4B and 4C). Among the 35 upregulated lncRNAs in OV, three ZFAS1, SMIM37, and LINC01614 showed statistically significant differences and were selected for further survival analysis (Supplementary Fig. 4D and 4E). Notably, high expression of ZFAS1 was significantly associated with poorer overall survival ($p = 0.0044$), highlighting its potential as a prognostic marker in ovarian cancer (Supplementary Fig. 4F).

Discussion

In this study, we conducted a comprehensive transcriptome analysis using TCGA-BRCA and TCGA-CESC datasets to identify dysregulated lncRNAs. Differential expression analysis revealed that a large number of lncRNAs were dysregulated in both cancers. Comparative analyses identified several shared dysregulated lncRNAs, indicating the presence of common oncogenic and tumor-suppressive mechanisms in BRCA and CESC. Our methodology to screen the lncRNAs repressed in both BRCA and CESC datasets identified lncRNAs well-documented in the literature for their tumor-suppressive roles such as MAGI2-AS3, PCAT19, TRIM52-AS1, and MEG3^{8,22-25}. Notably, downregulated lncRNAs, including ENSG00000267047.1 and FGF14-AS2, were found to be associated with tumor-suppressive pathways and correlated with better survival outcomes in BRCA patients.

Several upregulated lncRNAs identified in our screen, such as CRNDE and TMPO-AS1, are known in the literature for their oncogenic roles. For instance, CRNDE has been implicated in promoting tumor growth, metabolic reprogramming, and poor prognosis in cancers, while TMPO-AS1 is associated with enhanced cell proliferation and invasion^{26,27}. On the other hand, FAM83H-AS1 was significantly upregulated in BRCA and showed a strong association with advanced tumor stages, nodal involvement, and metastasis, suggesting its role as a potential prognostic biomarker. As shown in Figs. 5 and 6, FAM83H-AS1 expression was elevated in intermediate pathological stages and nodal involvement in BRCA, whereas in CESC, its expression was mainly linked to nodal metastasis rather than primary tumor progression. These findings suggest a context-dependent role of FAM83H-AS1 in different cancers, where its expression in BRCA is associated with both tumor aggressiveness and metastasis, whereas in CESC, it seems to play a bigger role in lymphatic spread. Kaplan–Meier survival analysis further confirmed that high FAM83H-AS1 expression was linked to poor survival outcomes in both BRCA and CESC patients, reinforcing its prognostic relevance. These findings indicate that dysregulated lncRNAs could serve as useful biomarkers for predicting prognosis and developing targeted therapies in gynaecological cancers.

Our findings also align with previous studies, where lncRNAs such as HOTAIR and MALAT1 were found to be strong predictors of survival and potential therapeutic targets in breast and cervical cancers^{12,28,29}. The expression patterns of these dysregulated lncRNAs suggest their involvement in key tumorigenic pathways, consistent with mechanisms reported for other well-known lncRNAs like MEG3, which activates p53 signaling and inhibits tumor progression^{8,9}. The presence of both shared and distinct dysregulated lncRNAs in BRCA and CESC suggests the existence of common yet cancer-specific regulatory mechanisms, emphasizing the need for more focused experimental validation to understand their precise roles. Furthermore, cytoplasmic-nuclear localization data obtained from lncAtlas provided additional mechanistic insights, linking nuclear localization to chromatin remodeling and transcriptional regulation, while cytoplasmic localization suggested roles in post-transcriptional modifications such as mRNA stability and translation²⁰. Our experimental validation in breast cancer cell lines supports the computational findings and reinforces the prognostic relevance of FAM83H-AS1 and FGF14-AS2 in relation to tumor aggressiveness.

Functional enrichment analysis using Enrichr and STRING provided further insights into the biological pathways associated with the dysregulated lncRNAs. Pathways such as chromatin organization and apoptosis were significantly enriched, emphasizing the diverse roles of these lncRNAs in tumor progression^{19,30}. For example, STRING analysis showed that FAM83H-AS1 was associated with oncogenic signaling, whereas ENSG00000267047.1 and FGF14-AS2 were linked to tumor-suppressive pathways. These findings are consistent with previous studies where HOTAIR and NEAT1 were found to regulate chromatin structure and cancer progression^{18,31,32}.

Moreover, identifying dysregulated lncRNAs offers an opportunity for personalized therapeutic approaches. For instance, the strong correlation between FAM83H-AS1 and poor survival outcomes suggests its potential as a therapeutic target, similar to how HOTAIR is being explored in preclinical models for targeted therapy¹². On the other hand, ENSG00000267047.1, which was linked to improved survival, could serve as a favourable prognostic biomarker, aiding in patient stratification for personalized treatment strategies.

Methodology

Data acquisition and processing

Transcriptomic datasets for TCGA-BRCA and TCGA-CESC were obtained from The Cancer Genome Atlas (TCGA) and analyzed using the TANRIC (The Atlas of Noncoding RNAs in Cancer) platform. TANRIC provides pre-processed lncRNA expression data for over 12,000 annotated lncRNAs, quantified as log₂ (RPKM + 0.001)¹. RPKM (Reads Per Kilobase of transcript per Million mapped reads) is a widely used normalization method that corrects for both transcript length and sequencing depth, enabling fair comparison of expression levels across samples. The addition of a small constant (0.001) prevents undefined values for transcripts with zero expression and reduces variance at low expression levels. This log-transformation improves data distribution and ensures interpretability for downstream statistical analysis. TANRIC applies a filtering threshold of RPKM ≥ 0.3 to retain only reliably expressed lncRNAs. The BRCA dataset consisted of 837 tumor and 105 normal samples, while the CESC dataset included 196 tumor and 3 normal samples.

Differential expression analysis was performed by computing the mean log₂-transformed expression of each lncRNA separately for the tumor and normal groups. Log₂ fold change (log₂FC) was calculated as the difference between these means, and the corresponding linear fold change (FC) was derived as $FC = 2^{\log_2 FC}$. A two-tailed unpaired Student's t-test was applied to the log-transformed expression values to determine statistical significance. lncRNAs with $FC \geq 2$ or ≤ 0.5 and $p < 0.05$ were considered significantly dysregulated. Clinical metadata, including pathological stage, lymph node involvement, metastasis status, and survival information, were retrieved from the Genomic Data Commons (GDC) portal. Kaplan–Meier survival analysis, log-rank tests, and Cox proportional hazards models were used to assess associations between lncRNA expression and clinical outcomes. All analyses were conducted using RStudio (version 2023.12.0 Build 369) and Microsoft Excel 2019.

Identification of dysregulated lncRNAs

Differentially expressed lncRNAs in BRCA and CESC were classified into upregulated and downregulated groups. A comparative analysis was carried out to identify shared and unique lncRNAs between the two cancers. The lncRNAs that showed high statistical significance and had known associations with cancer-related pathways were prioritized for further study. Among them, FAM83H-AS1, ENSG00000267047.1, and FGF14-AS2 were selected for their prognostic relevance and analyzed in greater detail.

Functional enrichment analysis

Functional enrichment analysis of the dysregulated lncRNAs was conducted using the Enrichr tool. Gene Ontology (GO) terms, disease networks, and tissue-specific regulatory elements associated with these lncRNAs were examined. Pathways related to tumor progression, including chromatin remodelling and apoptosis regulation, were emphasized. The analysis provided deeper insights into the molecular mechanisms by which these lncRNAs contribute to tumor development.

Cytoplasmic-Nuclear localization analysis

The cellular localization of selected lncRNAs was assessed using Cytoplasmic-Nuclear Relative Concentration Index (CN-RCI) data from lncATLAS. The localization patterns of FAM83H-AS1 were analyzed to understand their functional roles in tumor progression. The finding suggested that cytoplasmic localization was linked to post-transcriptional regulation. These insights helped in evaluating the potential therapeutic relevance of these lncRNAs. Visualization and Statistical Tools.

Survival analysis

Kaplan-Meier survival analysis was carried out to investigate the relationship between lncRNA expression levels and overall survival in BRCA and CESC patients. Patients were divided into high- and low-expression groups based on the median expression level of each lncRNA. The TANRIC platform was used to generate survival curves and integrate lncRNA expression data with clinical outcomes. Statistical significance was determined using the log-rank test ($p < 0.05$), which helped identify lncRNAs with potential prognostic relevance in gynaecological cancers.

Visualization and statistical tools

Heatmaps and box plots depicting lncRNA expression profiles were generated using RStudio 2023.12.0 Build 369. Venn diagrams were used to visualize shared lncRNAs between BRCA and CESC datasets. Statistical analysis and data visualization were performed using R, Microsoft Excel 2019, and PowerPoint 2019. Data were presented as mean $\pm$ standard deviation (SD) unless otherwise specified.

Cell culture

MCF10A (non-tumorigenic breast epithelial), MCF7 (less aggressive breast cancer), and MDA-MB-453 (aggressive triple-negative breast cancer) were maintained in Dulbecco's modified eagle's medium (DMEM) supplemented with 10% fetal bovine serum (FBS) along with 1% of 100 units/mL antibiotic and antimycotic solution at 37 °C in a humidified atmosphere with 5% CO₂.

RNA extraction and quantitative Real-time PCR

Total RNA was extracted from cells using TRIzol reagent (Takara Biosciences) and reverse transcribed into cDNA using Moloney murine leukemia virus reverse transcriptase (Invitrogen). The qRT-PCR reactions were carried out in duplicates in 10 μ L volume for the expression analysis. The reaction mixture contained SYBR Select master mix (2X, Takara Biosciences), cDNA template, and forward and reverse gene or lncRNA specific primers (0.1 μ M each). Target sequence amplification temperature profile followed was as follows: Initial denaturation for 10 min at 95 °C, followed by 40 cycles of 10 s at 95 °C and amplification for 30 s at annealing temperature of 60 °C. Finally, a melt curve analysis was carried out at a temperature range of 60 °C to 95 °C for 20 min. The GAPDH was used as an internal control for lncRNA and mRNA quantification. Results were calculated using the $\Delta\Delta C_t$ method to determine the fold change in expression between the experimental and control groups.

The RT-PCR primers are as follows:

FAM83H-AS1: ACTCACTGGGTCAGCAACAC; AAGCCCGGGTTGATATTGGG.FGF14-AS2: GTGGCG GATAAAGAGCCACT; CCTTCTCAAAGCGCCAACC.

Data availability

Transcriptomic analysis was conducted utilizing The Cancer Genome Atlas (TCGA) datasets for TCGA-BRCA and TCGA-CESC, obtained from the TANRIC database (https://ibl.mdanderson.org/tanric/_design/basic/download.html). Corresponding clinical data for this analysis were retrieved from the National Cancer Institute's Genomic Data Commons (GDC) data portal for TCGA-BRCA (<https://portal.gdc.cancer.gov/projects/TCGA-BRCA>) and TCGA-CESC (<https://portal.gdc.cancer.gov/projects/TCGA-CESC>).

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Author contributions

Madhur Sharma is the principal investigator of this study, who conceptualized the project, performed data analyses, and wrote the manuscript. M.S. designed the primary lncRNA screen that led to the identification of dys-regulated lncRNAs in TCGA-BRCA and TCGA-CESC datasets, including FAM83H-AS1, ENSG00000267047.1, and FGF14-AS2. M.S. conducted comparative cancer data analysis, functional enrichment studies, survival analyses, and cytoplasmic-nuclear localization experiments. N.C. and P.P. assisted with statistical analyses and visualization, while D.P.S., A.N., and S.S. provided supervision. All authors reviewed the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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